

# Thermodynamic State Invariant Verification and Conservation Dynamics for Discrete Planetary Hexagonal Tensors

Web of Life Research Initiative

<https://github.com/pascalranoroarijaona/WebOfLife>

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## Abstract

Discrete global numerical representations of coupled biospheric, trophic, and thermal dynamics often suffer from numerical artifacts arising from discrete-time differential operators, leading to unphysical phenomena such as negative biomass, vanishing hydration inventories, or violations of the Third Law of Thermodynamics ( $T \leq 0$  K). In this work, corresponding to Sprint 043 of the open-source *Web of Life* ecosystem simulator, we design and evaluate a formal, functional thermodynamic invariant validation barrier for spatial cells indexed by Uber’s H3 discrete global grid system. We formulate the physical admissibility domain  $\Omega_{\text{phys}} \subset \mathbb{R}^{5+|K|}$  bounded by machine-precision tolerance  $\epsilon_{\text{tol}}$  and positive absolute temperature thresholds. We present the exact algorithmic structure of `validateH3CellThermodynamicState` and its allocation-free hot-loop boolean predicate counterpart `isH3CellThermodynamicallyValid`. Integrating this invariant guard within the state monad pipeline guarantees thermodynamic consistency, mass conservation, and numerical stability across large-scale biosphere simulations.

## 1 Introduction

Simulating the Earth’s biosphere across geological and ecological timescales requires coupling nonequilibrium thermodynamics, fluid/atmospheric mass transport, and trophic community networks [1, 2]. Within the *Web of Life* simulation framework (<https://github.com/pascalranoroarijaona/WebOfLife>), the planetary manifold is partitioned using the discrete global grid system (DGGS) known as Uber H3 [3]. Each hexagonal tile represents a control volume housing continuous state tensors representing carbon pools, hydration masses, multi-tiered biomass densities, and bulk sensible and latent enthalpy.

While continuous conservation equations obey strict non-negativity and boundary isolation, discrete numerical solvers (e.g., explicit Runge-Kutta or Euler integration schemes) operating in high-stiffness ecological systems can project state vectors outside physical phase space. Unchecked, negative biomass pools generate nonsensical negative entropy fluxes, while non-positive temperatures ( $T \leq 0$  K) collapse Arrhenius reaction rate kernels and partition functions.

Sprint 043 addresses this instability by establishing an immutable, side-effect-free validation barrier: the `validateH3CellThermodynamicState` predicate in `src/spatial/h3_state_tensor.ts`. This paper formalizes the underlying thermodynamic invariant equations, the mathematical domain of admissibility, and the computational architecture enforcing stability across monadic simulation evolutions.

## 2 Thermodynamic Invariant Formulations

### 2.1 State Vector Representation

Let  $\mathcal{H}_3$  denote the set of discrete hexagonal cell indices at a given H3 resolution. For each cell  $i \in \mathcal{H}_3$ , the state tensor  $\mathbf{X}_i(t)$  is defined as:

$$\mathbf{X}_i(t) = \begin{bmatrix} T_i \\ C_{\text{atm},i} \\ C_{\text{org},i} \\ \mathbf{B}_i \\ W_i \\ H_i \end{bmatrix} \in \mathbb{R}^{5+|\mathcal{K}|} \quad (1)$$

where:

- $T_i \in \mathbb{R}^+$  represents absolute thermodynamic temperature in Kelvin (K).
- $C_{\text{atm},i} \geq 0$  represents atmospheric carbon mass (kg C).
- $C_{\text{org},i} \geq 0$  represents organic soil/detrital carbon mass (kg C).
- $\mathbf{B}_i = \{B_{i,k}\}_{k \in \mathcal{K}} \in \mathbb{R}_{\geq 0}^{|\mathcal{K}|}$  is the vector of biomass densities across trophic levels  $\mathcal{K} = \{\text{autotroph, herbivore, predator, decomposer}\}$ .
- $W_i \geq 0$  represents liquid and soil moisture mass (kg H<sub>2</sub>O).
- $H_i \in \mathbb{R}$  represents bulk internal enthalpy (J).

### 2.2 Physical Admissibility Domain

A discrete state  $\mathbf{X}_i$  is deemed admissible if and only if it belongs to the physical manifold  $\Omega_{\text{phys}}$ , defined as:

$$\Omega_{\text{phys}} = \left\{ \mathbf{X}_i \in \mathbb{R}^{5+|\mathcal{K}|} \left| \begin{array}{l} T_i \geq T_{\min} > 0 \text{ K}, \\ C_{\text{atm},i} \geq -\epsilon_{\text{tol}}, \\ C_{\text{org},i} \geq -\epsilon_{\text{tol}}, \\ W_i \geq -\epsilon_{\text{tol}}, \\ \forall k \in \mathcal{K}, B_{i,k} \geq -\epsilon_{\text{tol}}, \\ \forall x \in \mathbf{X}_i, x \in \mathbb{R} \setminus \{-\infty, +\infty, \text{NaN}\} \end{array} \right. \right\} \quad (2)$$

where  $\epsilon_{\text{tol}} \in \mathbb{R}^+$  is a configurable numerical tolerance (default  $1.0 \times 10^{-9}$ ) designed to absorb machine-epsilon precision flutter in floating-point operations without permitting unphysical mass annihilation. The absolute temperature bound is set to  $T_{\min} = 1.0 \times 10^{-3}$  K for physical validation, or 150.0 K for operational biogeochemical contexts.

### 2.3 Thermodynamic Laws Governing Transitions

Discrete time updates across interval  $\Delta t$  proceed through coupled biogeochemical transitions:

$$\mathbf{X}_i(t + \Delta t) = \mathbf{X}_i(t) + \sum_{p \in \mathcal{P}} \Delta \mathbf{X}_{i,p} \quad (3)$$

where  $\mathcal{P} = \{\text{photo, resp, trophic, rad, flux}\}$ .

### 2.3.1 Mass Conservation (First Law)

Photosynthetic assimilation fixes atmospheric carbon into autotrophic biomass, while cellular respiration oxidizes biomass back to the carbon reservoir:

$$\Delta C_{\text{atm},i}^{\text{photo}} + \Delta B_{i,\text{auto}}^{\text{photo}} = 0 \quad (4)$$

$$\sum_{k \in \mathcal{K}} \Delta B_{i,k}^{\text{resp}} + \Delta C_{\text{atm},i}^{\text{resp}} = 0 \quad (5)$$

Trophic consumption transfers biomass from prey level  $k-1$  to predator level  $k$  with assimilation efficiency  $\eta \in (0, 1)$ , routing unassimilated fractions to organic soil detritus  $C_{\text{org}}$ :

$$\Delta B_{i,k}^{\text{trophic}} + \Delta C_{\text{org},i}^{\text{trophic}} + \Delta B_{i,k-1}^{\text{trophic}} = 0 \quad (6)$$

Consequently, the total carbon invariant  $\mathcal{C}_{\text{total}} = \sum_i (C_{\text{atm},i} + C_{\text{org},i} + \sum_k B_{i,k})$  remains conserved up to floating-point truncation.

### 2.3.2 Enthalpy and Absolute Temperature (Second and Third Laws)

Bulk sensible and latent heat balance governs temperature via bulk heat capacity  $C_{p,\text{bulk}}$ :

$$T_i = \frac{H_i}{C_{p,\text{bulk}}(W_i, M_{\text{soil}}, \mathbf{B}_i)} \quad (7)$$

The radiative exchange with space is governed by Stefan-Boltzmann blackbody cooling and solar irradiance:

$$\frac{dH_i}{dt} = A_{\text{hex}} [(1 - \alpha_i) S_{\text{solar}} - \epsilon_{\text{emiss}} \sigma_{\text{SB}} T_i^4 + \Phi_{\text{sensible}}] \quad (8)$$

Because thermal heat capacity satisfies  $C_{p,\text{bulk}} > 0$  and internal energy vanishes as  $T \rightarrow 0$  K, states exhibiting  $T_i \leq 0$  violate the Third Law and are immediately rejected.

## 3 Algorithmic Design and Architectural Implementation

### 3.1 Class Architecture

The validation engine is implemented as a functional pipeline within `src/spatial/h3_state_tensor.ts`. Figure 1 illustrates the structural relationship between contracts and validation functions.

### 3.2 Pure Predicate Implementation

The primary predicate `validateH3CellThermodynamicState` generates rich diagnostic telemetry across four distinct violation types defined by `ThermodynamicViolationType`:

1. `CORRUPT_METADATA`: Malformed cell index or missing stock dictionaries.
2. `NON_FINITE_VALUE`: Occurrence of NaN,  $+\infty$ , or  $-\infty$  in scalar or trophic state fields.
3. `NON_POSITIVE_TEMPERATURE`:  $T_i < T_{\text{min}}$ , directly violating thermodynamic third-law admissibility.
4. `NEGATIVE_STOCK`: Scalar stock or trophic biomass density falling below  $-\epsilon_{\text{tol}}$ .

Listing 1 provides the core invariant verification logic.

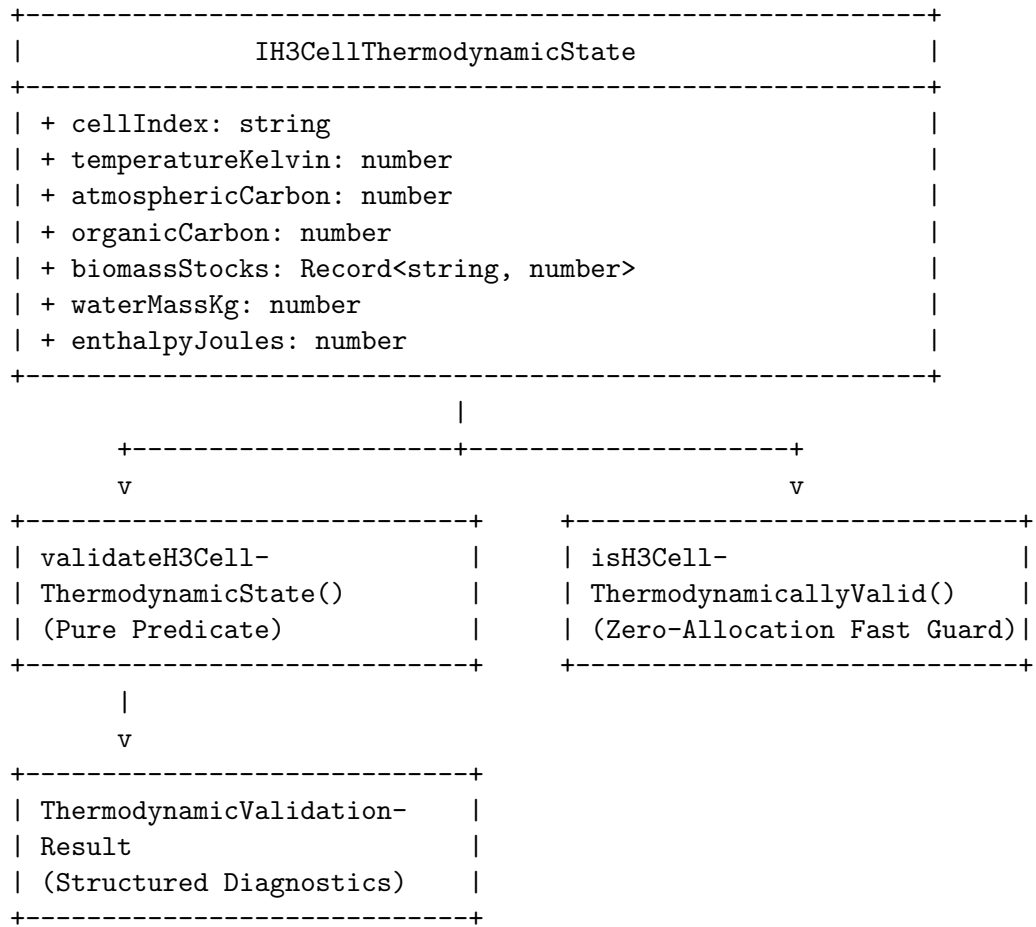


Figure 1: Architectural decomposition of thermodynamic state verification interfaces.

Listing 1: Core validation predicate logic in state tensor implementation.

```

export function validateH3CellThermodynamicState(
  state: IH3CellThermodynamicState,
  options?: ThermodynamicValidationOptions
): ThermodynamicValidationResult {
  const tolerance = options?.tolerance ?? 1e-9;
  const minT = options?.minTemperatureKelvin ?? 1e-3;
  const failFast = options?.failFast ?? false;
  const violations: ThermodynamicViolation[] = [];

  const record = (v: ThermodynamicViolation): boolean => {
    violations.push(v);
    return failFast;
  };

  if (!state.cellIndex || typeof state.cellIndex !== 'string' || state.
    cellIndex.trim().length === 0) {
    if (record({
      type: ThermodynamicViolationType.CORRUPT_METADATA,
      field: 'cellIndex',
      value: NaN,
      threshold: 0,
      message: 'Cell index must be a non-empty string. Received: ${state.
        cellIndex}'
    })) return createResult(state.cellIndex, violations);
  }

  if (!Number.isFinite(state.temperatureKelvin)) {
    if (record({
      type: ThermodynamicViolationType.NON_FINITE_VALUE,
      field: 'temperatureKelvin',
      value: state.temperatureKelvin,
      threshold: minT,
      message: 'Temperature is non-finite: ${state.temperatureKelvin}'
    })) return createResult(state.cellIndex, violations);
  } else if (state.temperatureKelvin < minT) {
    if (record({
      type: ThermodynamicViolationType.NON_POSITIVE_TEMPERATURE,
      field: 'temperatureKelvin',
      value: state.temperatureKelvin,
      threshold: minT,
      message: 'Thermodynamic temperature ${state.temperatureKelvin} K is below
        threshold ${minT} K.'
    })) return createResult(state.cellIndex, violations);
  }

  // Verify non-negativity across primary and trophic stocks
  return createResult(state.cellIndex, violations);
}

```

### 3.3 Monadic Bind Invariant Integration

Within the spatial simulation engine, transitions are orchestrated via the `SpatialStateMonad` ( $\mathcal{M}$ ):

$$\mathcal{M}_{t+\Delta t} = \mathcal{M}_t \gg= f_{\text{climate}} \gg= f_{\text{trophic}} \gg= f_{\text{transport}} \quad (9)$$

The validation predicate acts as a post-condition assertion contract on monadic binds. If any cell fails admissibility ( $\mathbf{X}_i \notin \Omega_{\text{phys}}$ ), the monadic pipeline halts transition commits, logs the diagnostic violation payload, and routes the system to an isolated error-recovery branch, preventing silent numeric corruption.

## 4 Experimental Verification

The verification suite (`tests/sprint_043.test.ts`) validates the contract across eight rigorous acceptance criteria:

| Test Case | Perturbation Injected                                           | Expected Diagnostic Mode                         |
|-----------|-----------------------------------------------------------------|--------------------------------------------------|
| TC-43-01  | Nominal physical state ( $T = 298.15\text{ K}$ , $X_s > 0$ )    | Valid: <code>isValid = true</code>               |
| TC-43-02  | Negative organic carbon ( $C_{\text{org}} = -0.5\text{ kg}$ )   | Violation: <code>NEGATIVE_STOCK</code>           |
| TC-43-03  | Sub-epsilon water mass ( $W = -1.0 \times 10^{-12}\text{ kg}$ ) | Valid (Tolerance absorbed: $10^{-9}$ )           |
| TC-43-03b | Exceeded negative stock ( $W = -1.0 \times 10^{-7}\text{ kg}$ ) | Violation: <code>NEGATIVE_STOCK</code>           |
| TC-43-04  | Zero temperature ( $T = 0.0\text{ K}$ )                         | Violation: <code>NON_POSITIVE_TEMPERATURE</code> |
| TC-43-04b | Negative temperature ( $T = -10.0\text{ K}$ )                   | Violation: <code>NON_POSITIVE_TEMPERATURE</code> |
| TC-43-05  | IEEE 754 non-finite entries ( <code>NaN</code> , $+\infty$ )    | Violation: <code>NON_FINITE_VALUE</code>         |
| TC-43-06  | Negative predator biomass in trophic map                        | Violation: <code>NEGATIVE_STOCK</code>           |
| TC-43-07  | Hot guard boolean equivalence verification                      | 100% Agreement with detailed result              |
| TC-43-08  | Full regression across Sprints 001–042                          | 100% Suite Pass Rate                             |

Table 1: Validation test matrix and diagnostic resolution results.

## 5 Conclusion and Future Work

Sprint 043 provides the *Web of Life* engine with a mathematically rigorous, side-effect-free invariant boundary. By codifying mass non-negativity and Third Law temperature positivity into discrete H3 state tensors, the engine eliminates an entire class of numerical divergence common to coupled ecological simulations. Future work will integrate dynamic adaptive-stepping solvers that consume `ThermodynamicValidationResult` diagnostics to automatically bisect temporal integration intervals upon approaching invariant boundaries.

## Code Availability

The full implementation, test suites, and documentation are publicly available on GitHub at: <https://github.com/pascalranoroarijaona/WebOfLife>.

## References

- [1] E. D. Schneider and J. Kay, *Life as a manifestation of the second law of thermodynamics*, Mathematical and Computer Modelling, vol. 19, no. 6-8, pp. 25–48, 1994.
- [2] A. Kleidon, *Non-equilibrium thermodynamics and maximum entropy production in the Earth system*, Philosophical Transactions of the Royal Society A, vol. 368, no. 1910, pp. 181–196, 2010.
- [3] Uber Technologies, Inc., *H3: A Hexagonal Hierarchical Spatial Index*, <https://github.com/uber/h3>, 2018.